

Edge-Capable Open-Weight LLMs in Modern Greek NLP: A Reproducible Benchmark

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Abstract

Natural Language Processing (NLP) for lesser-resourced languages faces persistent challenges, including fragmented datasets, inherited biases from high-resource settings, and limited reproducible evaluation protocols. This study addresses these gaps for Modern Greek through three contributions. First, we distill publicly available Greek datasets from prior survey work and classify them according to their suitability for benchmarking, pre-training, and retrieval-augmented generation. Second, we evaluate edge-capable open-weight large language models (LLMs) on seven Greek NLP task families for which data are available and legally reusable, revealing task-specific strengths and weaknesses across classification, sequence labeling, translation, correction, and summarization. Third, we provide a reproducible benchmark pipeline that connects dataset selection, model execution, metric reporting, publication-ready artifacts, and automated discovery of candidate datasets from newer literature. Together, these contributions support more transparent evaluation and reuse of Greek NLP resources.

CCS Concepts

• **Computing methodologies** → **Natural language processing**: Language resources; • **Information systems** → Data analytics.

Keywords

Greek NLP, benchmarking, large language models, reproducibility, evaluation

ACM Reference Format:

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1 Introduction

Natural Language Processing (NLP) has advanced substantially with the rise of large language models (LLMs), whose development

depends on immense volumes of digital data [5]. Although multilingual NLP has benefited from these advances, progress for lesser-resourced languages continues to lag behind that of well-supported languages. As a result, NLP for many of the world's languages often relies on methods, resources, and evaluation practices developed for high-resource settings, inheriting assumptions, biases, and other characteristics that may not align with the properties of less supported languages [2].

This gap is compounded by the scarcity of open-access, high-quality language resources. In this context, high-quality resources refer to native, human-authored corpora, as opposed to synthetic or machine-translated data. Although synthetic and translated resources are often used to bridge data gaps, they can introduce “translationese”, where the text inherits syntactic and stylistic constraints from the source language [16], and may lack the lexical diversity and morphological nuance of original language use. Such resources are needed both for pre-training foundation models and for building gold-standard benchmarks for downstream tasks. Comprehensive surveys of NLP resources for lesser-resourced languages therefore do more than document digital readiness: they can also guide future research, reveal reusable datasets, and support the construction of new benchmarks. Our work focuses on one such language, Modern Greek.¹

In this work, we ask two related research questions: *which properly licensed Greek datasets can be used to train or evaluate LLMs on downstream NLP tasks, and how do edge-capable open-weight LLMs perform across the resulting Greek benchmark?* By addressing these questions, our work makes three contributions, whose end-to-end pipeline is summarized in Figure 1:

- Based on a recent Greek NLP survey [2], we extract publicly available Greek datasets that are (a) open-access and (b) permit derivatives, aggregate them into a unified collection that is easy to recompile and use, and provide an automated discovery component for identifying candidate datasets from newer Greek NLP literature.²
- We classify the collected datasets according to their usability for benchmarking or pre-training.
- We benchmark edge-capable open-weight LLMs with manageable parameter counts and Greek-language capabilities across multiple NLP tasks, revealing task-specific strengths and weaknesses.

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¹In our study, the term “Greek” refers exclusively to Modern Greek (ISO 639-1: el), excluding previous stages of the language’s history.

²Our code and a leaderboard can be found at <https://doi.org/10.5281/zenodo.15314882>.

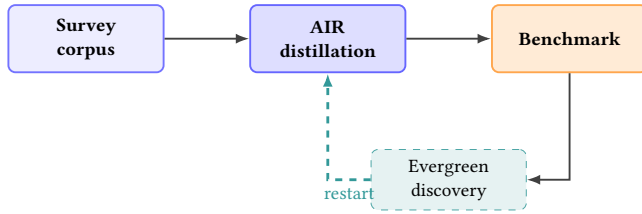


Figure 1: The benchmark pipeline. A survey corpus seeds AIR distillation (selecting accessible, interoperable, reusable datasets), which yields the benchmark. Evergreen discovery harvests newly published datasets and *restarts* the loop directly at distillation, bypassing the static survey corpus.

2 Related Work

2.1 Greek NLP Resources and Models

Modern Greek NLP has produced a sustained body of task-specific resources and systems over the past decades, including work on grammar checking, legal text processing, named entity recognition, sentiment analysis, and syntactic annotation [3, 11, 24, 29]. However, these resources remain unevenly distributed across tasks and domains, and that availability, licensing, and reusability are recurring barriers for reproducible experimentation [2]. This motivates our focus on datasets that are not only relevant to downstream Greek NLP, but also accessible and legally reusable.

Transformer-based models have also expanded the practical toolkit for Greek. Greek-specific pretrained models such as GreekBERT and GreekBART improved representation and sequence-to-sequence modelling for the language [9, 18], while recent open LLMs for Greek, such as Meltemi, indicate growing interest in language-specialised generative models [36]. In parallel, multilingual and open instruction-tuned LLMs such as Llama, Gemma, Aya, Qwen, and Falcon provide broader coverage and are increasingly used through local serving frameworks [1, 7, 13, 34, 40]. Our work complements these model-development efforts by evaluating manageable open models under a shared Greek benchmark protocol.

2.2 Benchmarking and Evaluation

Multi-task benchmarks such as GLUE and SuperGLUE established shared evaluation as a central mechanism for comparing general-purpose language understanding systems [37, 38]. Similar efforts have since appeared for individual languages, domains, and multilingual settings, including legal and lower-resource evaluation suites [6, 35, 39]. For foundation models and LLMs, broad evaluation frameworks such as HELM and the LM Evaluation Harness further emphasize reproducibility, task coverage, and transparent reporting [19].

Despite this progress, Greek LLM evaluation remains fragmented across datasets, task definitions, metrics, and execution environments. Existing Greek studies often target individual tasks or model families, making it difficult to compare open instruction-tuned LLMs across heterogeneous downstream settings. Our benchmark prioritizes a reproducible evaluation pipeline: it brings together properly licensed Greek datasets, standardizes task execution and

reporting, and produces outputs that can be inspected, reproduced, and reused in subsequent work.

3 The Data: From Collection to Distillation

Based on a systematic literature review covering 142 research NLP studies for Greek from 2012 to 2023 [2], we extracted and analysed information on 94 publicly available Greek datasets. Most of these resources were monolingual (84%), but bilingual and multilingual datasets were also present (i.e., with Greek being one of the languages). The most frequent domains were politics and news (20%) while the most frequent task was sentiment analysis (33%).

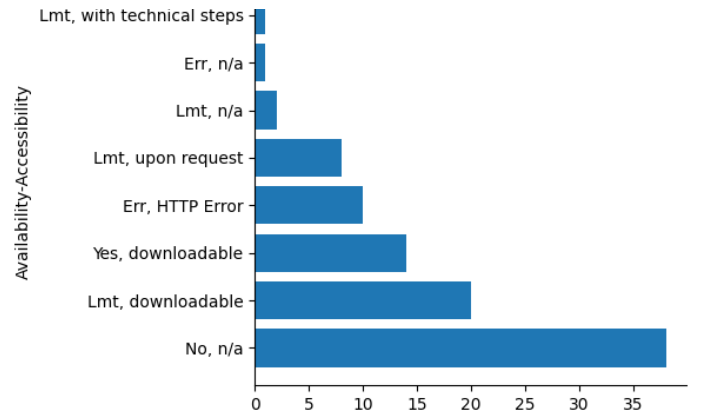


Figure 2: Availability and accessibility of collected datasets (tuple per bar). Availability is classified to Yes, Lmt (no license or otherwise constrained), Err (hindered by an error), No (not available via a URL). Accessibility is classified to downloadable, HTTP error, upon request, n/a (missing resource); i.e., the outcome we observed when accessing the resource.

3.1 Availability, Popularity and Licensing

Figure 2 shows the availability status of the collected datasets. The most common status is that **the datasets are not available**, meaning no information is provided by the authors of the respective study regarding their availability. The second most frequent category concerns datasets reported as downloadable, but **actually not publicly available** due to restrictions, such as missing license or subscription limitations. This limitation contradicts the principles and benefits of open access.

Studies with publicly available datasets, serving as a reference point, tend to attract more citations. This is shown by the higher third and fourth quartile of the green box in Figure 3. Furthermore, studies attracting citations while presenting data with limited [14] or no availability [12] belong to the past. The study of Juola and Stamatatos [14] is such an example, shown in Figure 4 with a blue peak in 2013: it concerns a shared task on authorship identification that attracted citations, but whose data lack a license.

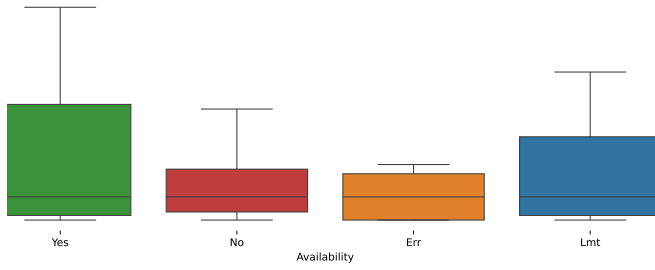


Figure 3: Whisker and box plots with the number of citations of studies per availability type.

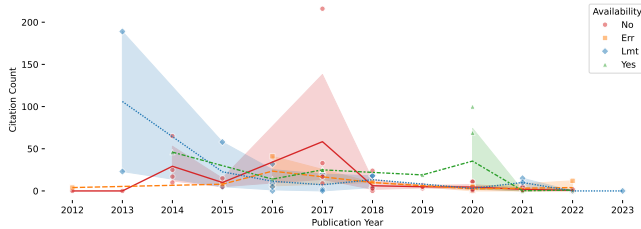


Figure 4: Citation counts per year of publication of studies developing datasets, based on the availability classified as yes, no, limited (lmt), or erroneous (err).

3.2 AIR Datasets Distillation

We collected accessible, interoperable and re-usable (AIR)³ datasets pooled from a recent survey [2]. We focused on the 14 Greek datasets that meet our criteria of being licensed, accessible, machine-actionable, and with no hidden costs. We excluded one [10] that was licensed under CC BY-NC-ND 4.0, which does not allow derivatives and thus limits reusability. We also excluded one [41] that shared tweets not via rehydration (i.e., only IDs shared), but as full texts.⁴ From the two resources released by Korre et al. [17], we retained the 227-sentence Greek Native Corpus (GNC) for GEC, but excluded the 100-sentence Greek WikiEdits corpus (GWE), which is too small for our study. We also excluded the dataset of open-access e-books [4], because it comprises OCR artifacts.⁵ Given these exclusions, we end up with ten datasets, presented in Table 1, which cover a variety of domains.

3.3 AIR Resources Suitable for RAG

Table 1 comprises three datasets with no ground truth or whose ground truth is automatically extracted (A/N; i.e., of uncertain

³To comply with the FAIR principles, datasets must have a unique and persistent digital identifier (commonly a Persistent Identifier/PID). A PID is a specific combination of numbers and letters that ensures datasets can be uniquely and sustainably accessed as digital objects over time. Merely providing a URL for accessibility does not make a dataset findable, as URLs can become obsolete or break if not properly maintained.

⁴Although it is possible that a company gained access to the texts (e.g., via firehose), releasing them with another license, a user should still be able to delete their content from all platforms. If not, then this dataset is not compliant with the rules. If so, then the dataset is not stable, harming reproducibility of experiments.

⁵Although typos and errors are not prohibited for experiments, we observe that the author names and the title was often injected in the main text, leading to data leaking for potential benchmarks.

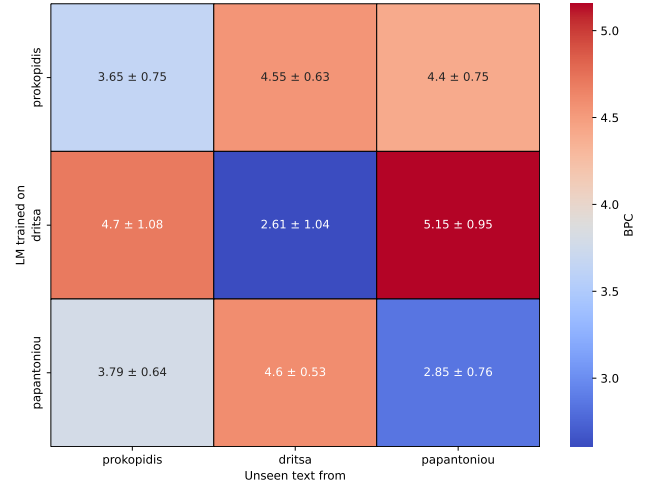


Figure 5: Heatmap of BPC measured on a sample per RAG dataset (average across the sample; shown horizontally) per trained LM (vertically); greater distance in warm colours.

quality). Although excluded from our benchmark, these datasets are still possibly useful for retrieval-augmented generation (RAG) and other purposes (e.g., pre-training). These concern 1.28M political speeches, spanning from July 1989 to July 2020 [8]; Greek Wikipedia event pages with named entity recognition (NER) and linking; 101,857 (cleaned) websites with open content [31], including online archives of Greek newspapers from 2003 to 2020, and the Greek part of the W2C corpus [21].

3.3.1 Dataset statistics. As shown in Table 2, the text length (measured in characters) greatly varies between the datasets. The dataset with the shortest texts on average is that of Papantoniou et al. [25] and that with the lengthiest is that of Prokopidis and Piperidis [31]. The dataset size (counted in number of files) also varies, starting from ten thousands [25], to one hundred thousands [31], to more than a million [8]. The aggregated text totals up to 1.4 billion characters and 211 million tokens (using white space split).

3.3.2 Linguistic distance. We investigate the linguistic associations between these datasets, using statistical language modelling (LM-ing) as a proxy. For each dataset, we trained one statistical character-level LM on the first 100 characters of 1,000 randomly selected texts per dataset.⁶ Then, we computed the Bits per Characters (BPC) across 500 randomly selected texts per dataset, reporting the mean. The result is displayed as a heatmap in Figure 5. High scores (in red) reflecting a greater linguistic distance between the dataset the LM is trained on and the one BPC is computed on. Low scores (in blue) reflect a low linguistic distance, with the lowest scores taking place in the diagonal, when predicting unseen text taken from the same dataset (self-distance). Based on this heatmap, we find that the dataset of Prokopidis and Piperidis [31] shows a higher self-distance compared to the rest while the datasets of Drita et al. [8]

⁶Character-level statistical LMing is an efficient way to capture stylistic features, employed in authorship analysis [26].

Table 1: Greek datasets with their annotation type (AT) defined as manual (M), automatic (A), curated (C; metadata provided by the distributor), hybrid (H; manual and automatic), user-generated (U; from user-edits, not curated) or no (N). We include also the host, size, size unit, domain and license of each dataset.

AUTHORS	AT	HOST	SIZE	SIZE UNIT	DOMAIN	LICENSE
Papantoniou et al. [25]	A	Zenodo	18,615	DOC	EVENTS	CC BY 3.0
Koniaris et al. [15]	C	Hugging Face	8,395	DOC	LEGAL	CC BY-NC 4.0
Rizou et al. [32]	M	msensis.com	2,176	SENT	ADMIN	CC-BY-NC-SA 4.0
Dritsa et al. [8]	A	Zenodo	1.28M	DOC	POLITIC	CC BY 4.0
Papaloukas et al. [24]	C	Hugging Face	47,563	DOC	LEGAL	CC BY 4.0
Korre et al. [17]	M	GitHub	227	SENT	ESSAYS	MIT
Bartziokas et al. [3]	H	GitHub	21,153	SENT	NEWS	CC-BY-NC-SA 4.0
Prokopidis and Piperidis [31]	N	nlp.ilsp.gr	101,857	DOC	WEB	CC-BY-NC-SA 4.0
Prokopidis and Papageorgiou [29]	H	GitHub	2,521	SENT	NEWS, POLITIC	CC BY-NC-SA 3.0
Prokopidis et al. [30]	U	nlp.ilsp.gr	17,018	DOC PAIR	NEWS	CC BY 4.0

Table 2: AIR datasets for RAG (not in the benchmark) with min, max, mean length in characters and size in documents.

	MIN	MAX	AVG	SIZE (#)
Papantoniou et al. [25]	34	1,484	172.9	18,615
Prokopidis and Piperidis [31]	36	67,963	2,430	101,857
Dritsa et al. [8]	1	291,949	900	1,280,927

and Papantoniou et al. [25] are far away from each other, especially when the LM is trained on the former and predicting the latter.

3.4 Resources Suitable for Downstream Tasks

The datasets used as benchmarks for downstream Greek NLP tasks are summarized in Table 3, along with key statistics and their associated supervised tasks. These resources cover three main categories: classification (Intent, Legal), sequence labeling (POS tagging, NER), and text-to-text generation (Grammatical Error Correction, Machine Translation, Summarization). Next, we present each one.

3.4.1 Intent Classification. Student queries, addressed to two university help desks, were manually annotated with six intent labels and three entity types [32]. This dataset is balanced across intent classes, with relatively short texts (average length: 58.6 characters). We use it to evaluate intent classification.

3.4.2 Legal Text Classification. A multi-level legal classification dataset is derived from the “Permanent Greek Legislation Code – Raptarchis” [24]. It includes hierarchical annotations of 47 volumes, 347 chapters, and 1,685 subjects. The dataset is highly imbalanced and exhibits large variation in document length (from 17 to approx. 1 million characters). The largest class comprises 34 examples (6.8%; science and arts) and the median has 10 examples. The smallest class has two examples (0.4%; media and tourism). We use it for volume classification.

3.4.3 POS Tagging. The Greek Universal Dependencies treebank was part of a project offering standardized treebanks with consistent annotations across languages [23]. This dataset [29] includes syntactic dependencies, POS tags, morphological features, and lemmas. It consists of 2,521 manually validated sentences split into train (1,622), development (403), and test (456) sets.

3.4.4 NER. The Greek NER dataset [3] comprises two annotation schemes: a coarse-grained set of 4 entity types (similar to CONLL-2003 [33]) and a fine-grained set of 18 types (similar to OntoNotes [28]). The dataset was developed during a Google Summer of Code project in 2018,⁷ where initial automatic annotation was followed by manual curation. The text length varies from 2 to 2,210 characters (Table 3). Both annotation levels exhibit heavy imbalance, with the four-tag level having an imbalance ratio (minor:major) of 0.02 and the 18-tag level showing near-zero imbalance. We assess NER at the less imbalanced four-entity-tag level.

3.4.5 Grammatical Error Correction (GEC). The Greek Native Corpus (GNC) [17] comprises (227) student-written sentences, annotated by human experts for (18) error types, related to grammar, spelling, punctuation, and morphology. Each sentence may contain zero, one, or multiple grammatical errors. We use this dataset to assess grammatical error correction. Because sentences with no annotated correction are uninformative for this task, we evaluate on the 175 sentences whose corrected form differs from the original (the remaining error-free sentences are excluded).

3.4.6 Machine Translation (MT). Bilingual corpora (756 language pairs) exist from content available in Global Voices, where volunteers translate news stories in 41 languages [30]. The 3,629 Greek documents are translated into 40 languages. However, not every document is translated into all languages, resulting in 17,018 bilingual document pairs involving Greek. We use the translations to English, Japanese, and Farsi, representing high-, moderately-, and low-supported language tiers [2].

⁷<https://github.com/eellak/gsoc2018-spacy>

Table 3: Dataset statistics for downstream Greek NLP tasks. **TEST:** capped test-set size used for evaluation; **UNIT:** instance granularity; **#CL.:** number of classes; **LEN.:** average text length in characters; **CIR:** class imbalance ratio (smallest-to-largest class proportion); **JSD:** Jensen–Shannon divergence (bits, 0=identical) between the capped and full test-set label distributions, computed over token-level tags for sequence-labeling tasks. Datasets split by us (fixed-seed stratified/random shuffle) are marked (†); the rest use the provider test split verbatim. CIR and JSD are undefined (–) for generation tasks; capping is inactive where TEST equals the full test size (intent, POS); for GEC, TEST is the 175 GNC sentences carrying at least one correction (of 227 total).

TASK	TEST	UNIT	#CL.	LEN.	CIR	JSD
Intent Cl. [32]	436†	sent	6	58.6	1.00	0.000
Legal Text Cl. [24]	500	text	47	3673.3	0.07	0.015
POS Tagging [29]	456	sent	16	138.2	0.03	0.000
NER [3]	500†	sent	4	180.9	0.02	0.000
GEC [17]	175	sent	–	483.0	–	–
MT [30]	1,500†	doc	3	125.1	–	–
Summarization [15]	300	text	–	18,771.3	–	–

3.4.7 Summarization. A legal corpus of 8,395 court decisions is compiled from the Greek Supreme Court (Areios Pagos),⁸ including the full texts, their summaries, case categories, and associated tags [15]. This dataset exhibits substantial length variability, with 18,541 characters on average and 18,351 st. dev. (min: 2,571; max: 303,538). We use it to assess summarization.

4 Benchmark Setup

4.1 Tasks and Models

Our benchmark covers seven task families: GEC, intent classification, legal text classification, machine translation, NER, POS tagging, and summarization. We evaluate six open-weight, instruction-tuned models in the 7B–9B parameter range that are available for local inference:

- qwen2.5:7b-instruct [40];⁹ hereafter called QWEN2.5.
- llama3.1:8b-instruct [13];¹⁰ hereafter called LLAMA3.1.
- llama-krikri-8B-Instruct;¹¹ hereafter called KRIKRI.
- aya-expanse:8b [7];¹² hereafter called AYA.
- gemma2:9b [34];¹³ hereafter called GEMMA2.
- falcon3:7b-instruct [1];¹⁴ hereafter called FALCON3.

This set controls for model scale while maximizing architectural and training diversity. LLAMA3.1 and QWEN2.5 provide long-context variants (128K), FALCON3 reports 32K context, and AYA reports 8K context. We include KRIKRI as a Greek-specialized extension of LLAMA3.1 to assess whether language-specific adaptation changes the task-level profile. Among Greek-adapted LLMs we evaluate

KRIKRI rather than the earlier Meltemi [36]. Besides being more recent, it continues LLAMA3.1 that is already part of our comparison, yielding a controlled base that isolates the effect of Greek adaptation. All models are evaluated under the same benchmark prompts, dataset splits, input truncation, decoding, and metrics, so observed differences mainly reflect model capability, multilingual transfer, and instruction-following quality rather than differences in the evaluation setup.

Inference setup. All models are served locally through Ollama,¹⁵ on a single commodity workstation.¹⁶ We use each model’s default Ollama build (4-bit Q4_K_M quantization), reflecting how an edge deployment would obtain it; every 7–9B model fits within the 11 GB VRAM. Decoding is greedy,¹⁷ which makes generation deterministic; any dataset-side sampling is fixed with seed 42. The maximum output length is 256 tokens (num_predict), which covers all task outputs, and each request has a timeout with up to three retries. We do not override the context window (num_ctx), so each model uses its default Ollama context. Summarization documents far exceed the other tasks in length (see Table 3), so their inputs are truncated to the median test-set length (13,408 characters) for all models alike; the remaining tasks fit the default context unmodified.

4.2 Deterministic Capped-Test Protocol

The full test sets are very uneven in size, with legal text classification and machine translation being especially large. To keep evaluation tractable while preserving reproducibility, we use a deterministic capped-test profile that always takes the first N instances in dataset order instead of drawing fresh random samples. The resulting capped evaluation (Table 3) contains:

- GEC: 175 examples
- Intent Classification: 436 examples
- Legal Text Classification: 500 examples
- Machine Translation: 500 English pairs, 500 Persian pairs, and 500 Japanese pairs
- NER: 500 examples
- POS Tagging: 456 examples
- Summarization: 300 examples

This design keeps the smaller tasks intact while reducing the run-time burden of the largest tasks. It also supports exact reruns because the selected subset is deterministic. For machine translation, the benchmark now uses a shared data-wrapper path rather than a separate task-local split implementation, which eliminates an earlier source of inconsistency in language-specific counts.

Provenance of splits. A natural concern is whether selecting the first N instances biases the class distribution. The answer depends on how each test set is ordered, and we use two regimes. For tasks where we construct the split ourselves (marked with † in Table 3), the test set is produced by a fixed-seed (42) stratified or random partition. Hence, taking the first N instances is equivalent to a deterministic pseudo-random sample. For tasks with an established

⁸<https://www.areiospagos.gr/>

⁹<https://huggingface.co/Qwen/Qwen2.5-7B-Instruct>

¹⁰<https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct>

¹¹<https://huggingface.co/ilspl/Llama-Krikri-8B-Instruct>

¹²<https://huggingface.co/CohereLabs/aya-expanse-8b>

¹³<https://huggingface.co/google/gemma-2-9b>

¹⁴<https://huggingface.co/tiiuae/Falcon3-7B-Instruct>

¹⁵HTTP /api/generate, non-streaming.

¹⁶An NVIDIA GeForce RTX 2080 Ti (11 GB VRAM), an Intel Core i7-9700K CPU, and 32 GB DDR4 RAM, on Ubuntu Server 24.04 LTS.

¹⁷Temperature 0 with default top_p/top_k.

provider split (legal text classification, POS, GEC, and summarization), we retain the original test set unchanged for comparability with prior work and take the first N instances in provider order. Either way, the cap preserves 100% of classes and the full label distribution: the Jensen–Shannon divergence to the full test set is at most 0.015 bits across all label-bearing tasks (JSD column, Table 3).

4.3 Metrics

We report task-appropriate primary metrics. For GEC, the primary metric is GLEU [22]; we also report exact match and edit-distance diagnostics (WER/CER) against reference and source text. For intent classification, legal text classification, NER, and POS tagging, macro-F1 is the primary comparison metric, with accuracy as a supplementary metric; for NER we additionally report entity-level F1. For machine translation we use chrF [27] as the primary metric and retain WER/CER as supplementary string-distance diagnostics. For summarization we use BERTScore-F1 [42] as the primary semantic comparison metric and report WER/CER and ROUGE [20] as supplementary overlap-based diagnostics. Latency is reported as average per-instance generation time in seconds. To compare models across heterogeneous metrics, we additionally compute a direction-aware normalized quality score per benchmark segment. Within each segment, the best model receives the highest normalized value and the worst model the lowest. These segment-normalized scores are then averaged to obtain an overall ranking.

5 Experimental Results

We analyse the experimental results by focusing on the best model per task, but also touching on the relationship between performance and efficiency. Table 4 reports the overall normalized ranking, and Table 5 summarizes the best model in each benchmark segment. Figure 6 visualizes normalized quality across segments, while Figure 7 shows the trade-off between quality and inference speed. Furthermore, an analytical table per task is available online.¹⁸

5.1 Patterns across models

5.1.1 The best model. The strongest overall model is GEMMA2, ranking first on the normalized aggregate with an average quality score of 77.6 and winning intent classification, NER, POS tagging, and two of the three machine translation segments.

5.1.2 The runner-up complements the best. LLAMA3.1 ranks second overall with an average normalized quality score of 61.0. It wins GEC and remains highly competitive in intent classification and summarization, but it trails GEMMA2 in the aggregate ranking.

5.1.3 Specialization matters. KRIKRI ranks fourth overall with an average normalized quality score of 48.5. It wins legal text classification and summarization, suggesting that Greek-specific adaptation helps on some tasks, but its summarization latency is substantially higher than the other models.

5.1.4 Competitive in cross-lingual tasks. QWEN2.5 ranks third overall and is especially competitive on Persian machine translation, where it wins the segment. AYA ranks fifth overall, but remains a

strong runner-up in summarization and in English and Japanese translation.

5.1.5 Ranking sensitivity. The ranking in Table 4 assigns equal weight to each of the nine evaluation segments. Since machine translation is evaluated on three language pairs, it contributes three segments to the overall ranking. Reweighting so that each of the seven task families counts equally (i.e., collapsing the three translation segments into a single averaged family) leaves the main conclusion intact: GEMMA2 stays first (80.3), LLAMA3.1 second (63.0), and FALCON3 last (19.8). A slight movement regards the tightly clustered middle group of QWEN2.5 (43.3), KRIKRI (45.5), and AYA (42.2), all within about two normalized points; i.e., a statistical tie rather than a reordering.

5.1.6 Prompt sensitivity. All benchmark prompts are zero-shot and identical across models. To probe sensitivity to prompt design, we re-ran intent classification (a fast task with a clean metric) under an alternative prompt, holding the split, decoding, and seed fixed: a semantic *paraphrase* that rewords the instructions but keeps the response format.¹⁹ Paraphrasing barely moves macro-F1: the mean absolute change is 0.05 and the top two (GEMMA2 and LLAMA3.1) keep their order. Hence, the conclusions are robust to wording.

5.2 Patterns across tasks

5.2.1 Text classification. Legal text classification remains a difficult benchmark component. As shown in Table 5, the best macro-F1 is 0.156 with KRIKRI, followed by 0.126 with GEMMA2.²⁰ This suggests that the task is still demanding under zero-shot prompting and may benefit from better label descriptions, exemplars, or task-specific fine-tuning. By contrast, intent classification is comparatively more tractable. As reported in Table 5, GEMMA2 achieves a macro-F1 score of 0.737, substantially outperforming other models on this task.

5.2.2 Translation. The capped profile evaluates 500 examples per target language, which is enough to separate models consistently while keeping runtime manageable. As summarized in Table 5, GEMMA2 achieves the highest chrF score for English and Japanese translation. Notably, its chrF score exceeds the rest of the field despite not achieving the best WER, indicating that chrF (character-level) and WER (word-level) capture different aspects of translation quality and can rank models differently. QWEN2.5 clearly outperforms other models on Persian translation, indicating that some models are better equipped to handle lower-resource or typologically distant language pairs such as Greek–Persian.

5.2.3 POS tagging and NER. As shown in Table 5, GEMMA2 leads on macro-F1 in both POS tagging and NER. However, LLAMA3.1 achieves higher NER accuracy, suggesting differences in handling minority entity classes and class imbalance.

5.2.4 Summarization. Under a common input truncation for all models, KRIKRI wins summarization with a BERTScore-F1 of 0.533, followed by AYA (0.519) and LLAMA3.1 (0.517). This suggests a semantic gain for the Greek-specialized model, although its average latency is markedly higher at 33 seconds per example.

¹⁸See the leaderboard at <https://doi.org/10.5281/zenodo.15314882>

¹⁹Prompts are shared in Appendix B, along with the results (Table 6 of the Appendix).

²⁰We replaced opaque identifiers with coarse-grained volume labels.

Table 4: Overall ranking of the evaluated models across the nine benchmark segments in the deterministic capped benchmark run. Normalized quality is averaged over segment-level, direction-aware scores.

Model	Rank	Avg. Quality	Avg. Latency (s)	Segment Wins
GEMMA2	1	77.6	1.50	5
LLAMA3.1	2	61.0	1.16	1
QWEN2.5	3	49.1	1.40	1
KRIKRI	4	48.5	11.96	2
AYA	5	48.0	1.14	0
FALCON3	6	15.9	1.34	0

Table 5: Best-performing model per benchmark segment on the deterministic capped evaluation run (GEC=175, Intent=436, Legal=500, MT(ENG=500, FAS=500, JPN=500), NER=500, POS=456, Summarization=300), using representative metrics per task.

Task Segment	Winner	Primary Metric	Value
GEC	LLAMA3.1	GLEU	0.796
Intent	GEMMA2	Macro-F1	0.737
Legal	KRIKRI	Macro-F1	0.156
MT-ENG	GEMMA2	chrF	61.781
MT-FAS	QWEN2.5	chrF	18.325
MT-JPN	GEMMA2	chrF	16.726
NER	GEMMA2	Macro-F1	0.122
POS	GEMMA2	Macro-F1	0.486
Summ.	KRIKRI	BERTScore-F1	0.533

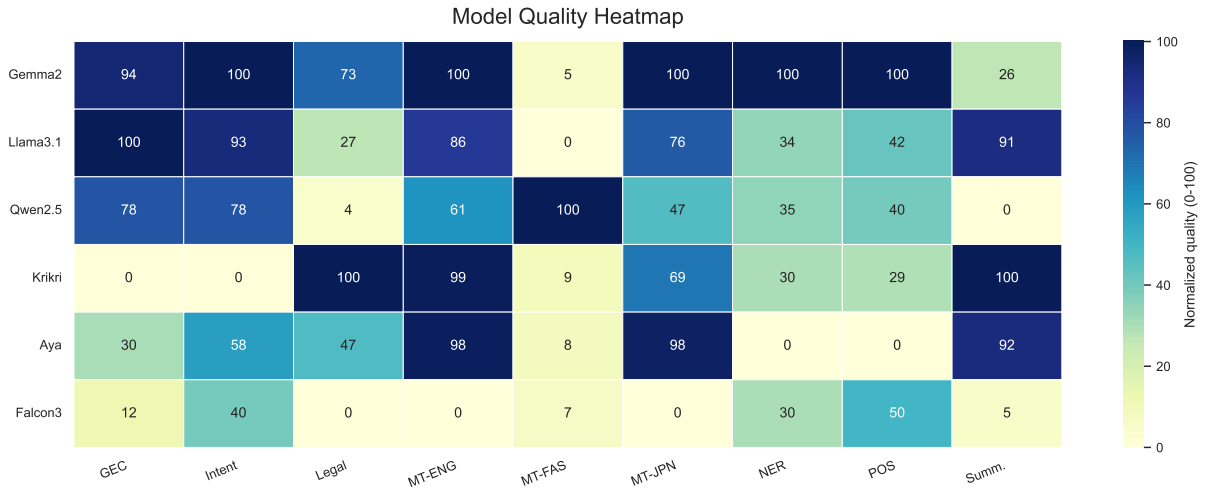


Figure 6: Normalized quality heatmap across the nine benchmark segments. Scores are normalized within each segment so that higher is always better, even when the underlying metric is lower-is-better such as WER.

6 Discussion

The benchmark combines multiple Greek NLP settings in a single reproducible workflow, but the current evaluation still has limitations. The capped protocol is intentionally pragmatic rather than exhaustive. Although well suited for iterative comparison, it is not a substitute for full test-set evaluation when final archival numbers are required. Specific metric choices are also provisional. In

particular, summarization and machine translation would benefit from stronger semantic metrics and human assessment, while legal text classification would benefit from richer label semantics. Even so, the supplemental metrics improve interpretability relative to a WER-only or macro-F1-only summary.

Despite these caveats, the benchmark exposes meaningful structure in model behaviour. Some models offer strong general performance across multiple tasks, while others are better understood as

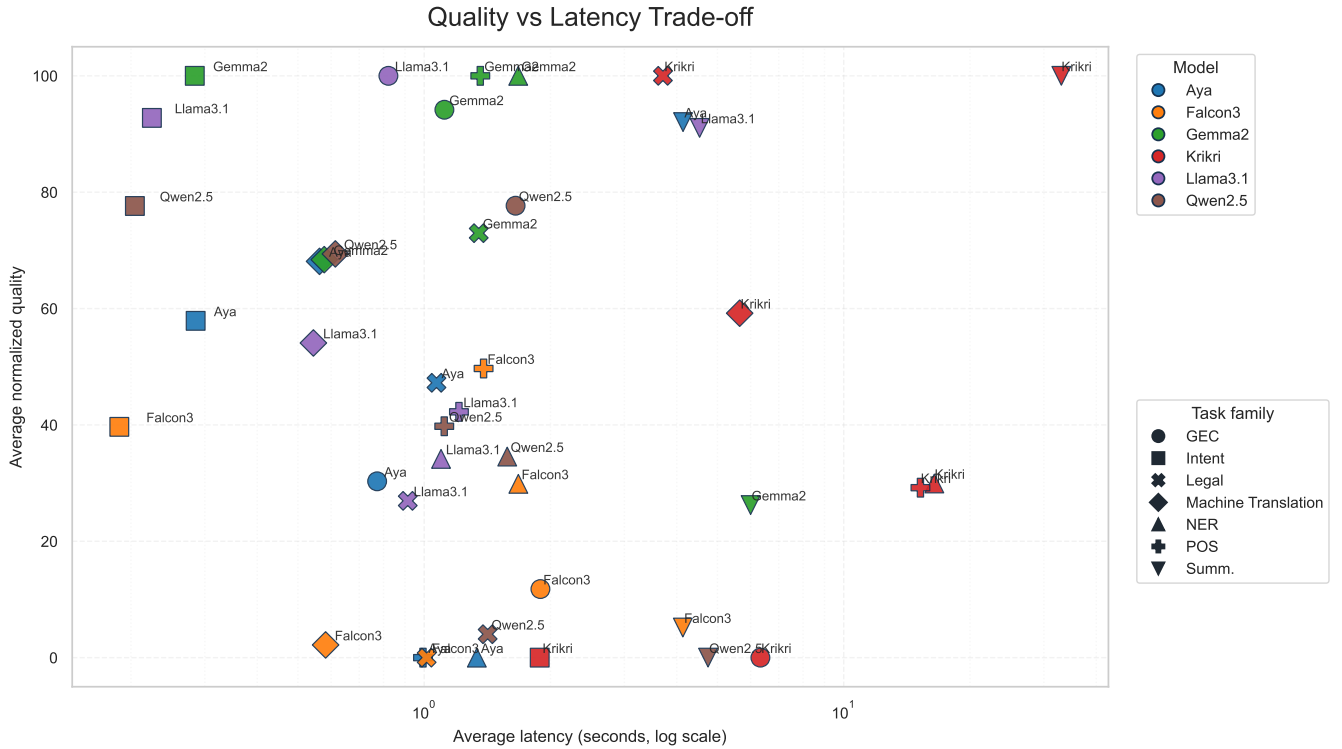


Figure 7: Quality-latency trade-off across models and task families. Each point is a model-task pair after averaging within the task family. The horizontal latency axis is log-scaled (slower summ. runs remain visible without compressing faster models).

specialists for translation or summarization. This kind of cross-task perspective is hard to obtain from isolated task runs and is exactly where a unified benchmark becomes valuable.

The two wins of KRIKRI are both legal-domain tasks. This is consistent with its Greek-focused continual pre-training and documented domain-oriented post-training for legal and long-document applications, although the public documentation does not let us verify whether benchmark-specific or closely related data were included in training.

7 Evergreen Dataset Discovery

Beyond our static benchmark, Greek NLP resources continue to evolve. To keep the resource collection maintainable rather than fixed to the 2012–2023 survey, we implement a lightweight *evergreen* discovery pipeline for newly published work adapting the search and screening protocol of Bakagianni et al. [2] to recent literature (2024–2026) retrieved from the *ACL Anthology* and *Semantic Scholar* ($n = 118$).²¹ It is a proof of concept that demonstrates how the resource inventory can be updated as new datasets appear, and does not affect the benchmark scores reported above.

Screening applied two stages: rule-based filtering removed non peer-reviewed publications and a zero-shot LLM classifier handled the remaining exclusion criteria. This yielded 55 in-scope papers, 48 showed abstract-level indications of a released dataset. For these

papers, dataset URLs were recovered automatically through a retrieval pipeline: article text was extracted, URLs were identified with regular expressions, and *Meta-Llama-3-70B*²² selected the most plausible dataset link. This produced 18 candidate dataset URLs, 13 of which were verified as reachable.

A manual audit on a small dataset-annotated subset ($n = 18$) showed high precision and moderate recall (precision = 1.00, recall = 0.69), a conservative but reliable behavior. Finally, titles and abstracts were classified into canonical_task categories using again *Meta-Llama-3-70B*, organizing newly identified resources (mainly text classification and machine translation) by task. These results indicate that automatic mining can effectively complement manual curation to enrich Greek NLP resources.

Licensing and integration. As Figure 1 shows, the evergreen loop restarts at AIR distillation rather than at the benchmark. Hence, every discovered candidate passes through the same accessibility (and licensing-aware) screening as the original pool. That is, only open-access resources that permit derivatives are retained, and promoted to a new benchmark segment in a future version only if they additionally expose a usable test split with an established metric. Combined with the conservative, precision-favoring audit above, this gives a license-safe path for the inventory to grow while keeping the current benchmark fixed and reproducible.

²¹The implementation is available in our GitHub repository, enabling reuse and continuous application of the protocol for future dataset discovery.

²²<https://huggingface.co/meta-llama/Meta-Llama-3-70B>

8 Conclusion

We presented a distilled collection of legally reusable Greek datasets; a benchmark over seven task families; and an end-to-end pipeline that links dataset selection, model execution, metric reporting, and evergreen dataset discovery. Using this benchmark we compared six edge-capable (quantized) open-weight LLMs across nine segments. GEMMA2 ranks first overall, while LLAMA3.1, QWEN2.5, and KRIKRI emerge as specialists on GEC, Persian translation, and Greek-heavy legal classification and summarization respectively. The results also show that some benchmark components remain difficult under zero-shot prompting, especially legal text classification. Finally, the evergreen discovery experiment indicates that automatic screening can complement manual curation by surfacing reachable candidate dataset links from recent Greek NLP literature. Future work should extend the benchmark with richer label semantics for legal text classification, stronger semantic metrics and human assessment for translation and summarization, and continuous dataset discovery to keep the resource collection up to date.

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A Prompt Templates

System (S) and user (U) prompt templates are listed below. Last is presented the prompt we used to investigate prompt sensitivity (intent classification, paraphrased), discussed next.

Grammatical Error Correction (GEC)

- S** You are a careful grammatical error correction system for Modern Greek. Correct the user’s sentence while preserving its meaning. Return only the corrected Greek sentence and nothing else.
- U** Correct the following Greek text grammatically and orthographically. Preserve the same meaning. Text: <ORIGINAL_TEXT>. Corrected text:

Legal Text Classification

- S** You are a Greek legal text classification system. Return only one label from the provided label set and nothing else.
- U** Given the following Greek legal text, return only its coarse-grained legal volume label. Choose exactly one label from: <LABEL_SPACE>. Text: <TEXT>. Label:

Machine Translation

- S** You are a translation system. Translate the user’s text accurately. Return only the translation and nothing else.
- U** Given a text in Greek, translate it to <TARGET_LANG>. Only the translation should be returned. Text: <SOURCE_TEXT>. Translation:

Named Entity Recognition

- S** You are a Greek named entity recognition system. Return only a Python-style list of tags and nothing else.
- U** You are given a tokenized Greek sentence. Return only a Python-style list of NER tags, one tag per token. Use only these labels: <NER_LABEL_LIST>. Tokens: <TOKEN_LIST>. Tags:

POS Tagging

- S** You are a Greek POS tagging system. Return only a Python-style list of tags and nothing else.
- U** You are given a tokenized Greek sentence. Return only a Python-style list of POS tags, one tag per token. Use only these labels: <POS_LABEL_LIST>. Tokens: <TOKEN_LIST>. Tags:

Summarization

- S** You are a Greek summarization system. Produce a concise summary in Greek and return only the summary.
- U** Given a Greek legal text, provide its summary also in Greek. Only the summary should be returned. Text: <DOCUMENT_TEXT>. Summary:

Intent Classification

- S** You are a Greek intent classification system. Return only the intent label and nothing else.
- U** Given a text, provide the intent of the text. Only the intent should be returned. Possible intents: <LABEL_SPACE>. Text: <TEXT>. Intent:

Intent Classification, paraphrased

- S** You classify the intent behind Greek user messages. Respond with exactly one intent label and no other words.
- U** Read the Greek text and identify its intent. Reply with a single label only. Available intents: <LABEL_SPACE>. Text: <TEXT>. Intent:

B Prompt Sensitivity

We re-ran the fast task of intent classification under the benchmark prompt (BASE) and a zero-shot alternative, holding the split, decoding (greedy, temperature 0), and seed (42) fixed so that only the prompt varies. PARA is a semantic paraphrase that rewords the instructions while keeping the response format (see Appendix A). The answer is kept to a bare intent label, so the task’s exact-match label normaliser applies unchanged. As a reproducibility check, the BASE column reproduces the main-text intent macro-F1 (e.g. GEMMA2 0.736 here vs. 0.737 in Table 5). Table 6 shows that para-

Table 6: Prompt sensitivity of intent-classification macro-F1. BASE: benchmark prompt; PARA: semantic paraphrase; Δ is relative to BASE.

MODEL	BASE	PARA	Δ
GEMMA2	0.736	0.730	−0.006
LLAMA3.1	0.707	0.723	+0.016
QWEN2.5	0.645	0.642	−0.003
AYA	0.564	0.505	−0.058
FALCON3	0.488	0.451	−0.037
KRIKRI	0.326	0.480	+0.154
MEAN $ \Delta $			0.046

phrasing barely moves the scores: the mean absolute change is 0.05, and the top two models keep their order. Future work could expand this test to investigate also reformatting.